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THE USE OF PRELIMINARY DATA IN ECONOMETRIC FORECASTING

E. Philip Howrey*

I. Introduction

TIMELY macroeconomic forecasts rely on preliminary data as well as on data in various stages of revision. In the specification and estimation of econometric models, however, the distinction between preliminary and revised data is usually ignored. This results in models that are not explicitly designed to use the data that are available for ex ante forecasts and leads to unnecessarily large forecast errors.

The existence of measurement error in macroeconomic data has been widely recognized. Previous studies have been concerned with three aspects of measurement error. The early studies by Zellner (1958) and Morgenstern (1963) were predominantly descriptive analyses of the relationship between preliminary and revised data. More recently, the sensitivity of parameter estimates to data revision has been investigated by Cole (1969), Denton and Kuiper (1965), Denton and Oksanen (1972), and McDonald (1976). Finally, the loss in forecast accuracy resulting from the use of preliminary rather than revised data has been studied by Cole (1969a, b), Denton and Kuiper (1965), and Stekler (1967).

It is generally agreed that the availability of more accurate preliminary data would result in better forecasts. Indeed, Cole (1969, p. 81) concluded that "the use of preliminary rather than revised data led to a *doubling* of the forecast errors." All of these previous studies may overstate the potential gain in forecast accuracy attributable to an improvement in preliminary data, however. The reason for this is that in these past studies the forecast procedures that have been evaluated do not make efficient use of preliminary data. In particular, the simple fact that the preliminary data are subsequently revised has been overlooked and this omission unnecessarily inflates the variance of the forecast errors.

This article outlines a general approach to the efficient use of preliminary data in econometric forecasting. Under certain conditions, which are summarized in the next section, an optimal predictor based on preliminary data can be derived. Two relatively simple forecasting models are examined to indicate how much can be gained through the use of optimal predictors over more traditional, suboptimal prediction methods. The first model is a simple autoregressive process applied to real disposable income. The second model is a two-equation system consisting of the disposable income equation and a simple consumption function. A byproduct of this analysis is an indication of the way in which forecast accuracy varies with the length of the delay in the availability of revised data.

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II. The Optimal Use of Data with Measurement Error

A general approach to the optimal use of error-infested data in forecasting is readily available; namely, the so-called Kalman filter approach. This solution to the prediction problem is briefly summarized, and the extensions and modifications that are required to adapt the solution to econometric forecasting are then considered.

It is assumed that the dynamic forecasting model can be written as

$$x_t = \phi x_{t-1} + u_t \tag{1}$$

where x_t is a vector of n endogenous variables, u_t is a vector of random disturbances, and ϕ is a matrix of known coefficients. The model can easily be generalized to include a vector of exogenous variables and more than a single lagged value of the vector of endogenous variables. It is frequently implicitly assumed that the endogenous variables are observed without error. Here, however, it is assumed that the observed values are related to the actual (unobserved) variables by an observation model of the form

$$y_t = \theta x_t + v_t \tag{2}$$

where y_t is a vector of m observed values, θ is an $m \times n$ matrix of known coefficients, and v_t is a vector of random measurement errors. In the limiting case, $\theta = I$ and $v_t = 0$ so that the endogenous variables are actually observed without error. It is not necessary that θ be a diagonal matrix nor is it necessary that it be a square matrix. Thus, it is possible to accommodate situations in which the observations consist of linear combinations of the endogenous variables and cases in which some of the endogenous variables are completely unobserved.

The one-period prediction problem for this system is to predict the endogenous variables x_{t+1} on the basis of the information that is available at time t. This information set consists of the current and previous observations, which will be denoted by $Y_t = (y_t, y_{t-1}, \ldots, y_1)$. Under certain conditions the optimal predictor is the conditional expectation of x_{t+1} given Y_t, i.e.,

$$\hat{x}_{t+1} = E[x_{t+1} \mid Y_t]. \tag{3}$$

In general, Astrom (1970, pp. 213–214) shows that the conditional expectation is optimal in the sense that it minimizes expected loss if (i) the loss function is symmetric and nondecreasing for positive arguments, and (ii) the conditional distribution of x_{t+1} given Y_t has a unimodal density function that is symmetric about $\hat{x}_{t+1}$.

On the assumption that u_t and v_t are mutually uncorrelated, serially independent multivariate normal processes and the loss function is symmetric, the conditional expectation is the optimal predictor. If $u_t \sim N(0, \Sigma_u)$ and $v_t \sim N(0, \Sigma_v)$, a recursive expression for the conditional expectation of x_{t+1} given $y(t)$, $y(t-1), \ldots, y(1)$ is[1]

$$\hat{x}_{t+1} = \phi \hat{x}_t + K_t(y_t - \theta \hat{x}_t) \tag{4}$$

where

$$K_t = \phi P_t \theta' (\theta P_t \theta' + \Sigma_v)^{-1} \tag{5}$$

$$P_{t+1} = (\phi - K_t \theta) P_t \phi' + \Sigma_u \tag{6}$$

with the initial conditions

$$\hat{x}_1 = 0 \tag{7}$$

$$P_1 = V[x_1 - \hat{x}_1]. \tag{8}$$

The computation of the optimal forecast proceeds recursively. In order to initiate the forecast process, an initial forecast and the variance of the initial forecast error are required. The optimal forecast of x_1 given no observations is simply the unconditional expectation of x_1. Using (1), it is not difficult to verify that $x_1 \sim N(0, \Sigma_x)$ where Σ_x is the solution of

$$\Sigma_x = \phi \Sigma_x \phi' + \Sigma_u. \tag{9}$$

Therefore the unconditional expectation of x_1 is zero and the variance P_1 of the forecast error is Σ_x. During the first few periods the covariance matrix of the forecast errors, P_t, and the weights, K_t, in the forecast equation will vary with t. Under certain conditions these matrices will converge to constant values.

The prediction problem posed by preliminary data is a special case of the measurement-error prediction problem. Certain modifications of the general approach are required to deal with economic data. The major changes involve the observation model.

National Accounts data are published quarterly in the *Survey of Current Business* by the Office of Business Economics. Since 1965, a preliminary estimate has been published one month after a quarter has ended (i.e., January, April, July, and October). The following month a first revision is published. In July of each year, the figures for the first quarter of the current year as well as the previous three years are revised. The first revision of the data is typically negligible and will not be discussed further here; the July revisions are much more important. Thus there is a lag of at least one quarter and at most four quarters before revised data are available. Moreover, routine revisions continue to be made for three years. In addition to these routine revisions, major benchmark revisions are periodically published.

[1] For a detailed derivation and explanation of this result, the reader is referred to Astrom (1970, pp. 218–230).

These benchmark revisions often involve conceptual changes and will not be dealt with here.

The sequence of routine revisions of national accounts data can be formalized as an observation system in the following way. The r^{th} observation on x_t is denoted by $y_{r,t+l(r)}$ to designate the fact that it is available with a lag of $l(r)$ periods. The observation system is then written as

$$y_{rt} = \theta_r x_{t-l(r)} + v_{rt} \qquad (r = 1, 2, \ldots, R). \tag{10}$$

In actual fact, $l(r)$ is a periodic function of t since the lag in availability of revised data depends on the quarter which is being considered. This time subscript is suppressed for notational convenience.

In principle, it is possible to consider the full set of observation equations for $R=4$. This corresponds to the preliminary estimates y_{1t} and the three subsequent July revisions (y_{2t}, y_{3t}, y_{4t}). However, for expository purposes, it is sufficient to consider the simplified system consisting of two equations:

$$y_t = \theta x_t + v_t \tag{11}$$

$$z_t = x_{t-l}. \tag{12}$$

The first of these equations describes the relationship between the preliminary estimate and the actual value. The second equation states that the first revision is identically equal to the actual value but is available only after l periods have elapsed. Although this is admittedly an unrealistic assumption, it is instructive to begin with this simple formulation in this initial investigation.

Realism demands one further modification of the observation system. Previous investigators have found that data revisions exhibit significant serial correlation. The empirical work summarized below suggests that a model of the form

$$v_t = \psi v_{t-1} + w_t \tag{13}$$

is sufficient to characterize the serial correlation of revisions. The disturbance process w_t is assumed to be serially uncorrelated. This completes the description of the observation system which consists of equations (11), (12), and (13).

Computation of the optimal predictor for the forecasting model (1) and the observation system (11)–(13) is simplified by rewriting the system. The observation system can be rewritten as

$$\begin{bmatrix} y_t \\ z_t \end{bmatrix} = \begin{bmatrix} \theta & 0 & \cdots & 0 & I \\ 0 & 0 & \cdots & I & 0 \end{bmatrix} \begin{bmatrix} x_t \\ x_{t-1} \\ \vdots \\ x_{t-l} \\ v_t \end{bmatrix}$$

or

$$Y_t = \Theta X_t. \tag{14}$$

Combining (1) and (13), the (revised) forecasting model can be expressed as

$$\begin{bmatrix} x_t \\ x_{t-1} \\ \vdots \\ x_{t-l} \\ v_t \end{bmatrix} = \begin{bmatrix} \phi & 0 & \cdots & 0 & 0 & 0 \\ I & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & I & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & \psi \end{bmatrix} \times \begin{bmatrix} x_{t-1} \\ x_{t-2} \\ \vdots \\ x_{t-l-1} \\ v_{t-1} \end{bmatrix} + \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \\ w_t \end{bmatrix}$$

or

$$X_t = \Phi X_{t-1} + U_t. \tag{15}$$

The system consisting of equations (14) and (15) is of precisely the same form as (1) and (2) so with an obvious change in notation, the solution given in (4) is valid.

III. Forecasts of Disposable Income

It is instructive to compare the optimal predictor developed in the previous section with several suboptimal predictors. This will be done first for a simple autoregressive process applied to real disposable income. The optimal predictor is derived and compared with three suboptimal predictors. The first suboptimal predictor ignores preliminary data entirely; the second uses preliminary data as if they were measured without error;[2] and the third adjusts the preliminary data for bias and serial correlation. The forecast accuracy of these alternative methods is investigated for forecasts of both preliminary and revised data.

[2] This predictor is of special interest because it appears to correspond closely to current practice in econometric forecasting.

The Forecasting and Observation Models for Disposable Income

The forecasting equation that is used in the prediction of real disposable income is a fourth order autoregressive process in first differences. This specification was chosen on the basis of an examination of the autocorrelation and partial autocorrelation functions of first differences of the most recently published values of real disposable income for the period 1954:I through 1974:IV. Parameter estimates and t-statistics, shown below, were then obtained from a stepwise autoregression of Δx_t on $\Delta x_{t-1},\ldots,\Delta x_{t-6}$. The result of the stepwise estimation procedure is

$$\begin{aligned}\Delta x_t = \underset{(2.33)}{1.920} + \underset{(4.61)}{0.478}\,\Delta x_{t-1} + \underset{(2.69)}{0.309}\,\Delta x_{t-3}\\ - \underset{(2.27)}{0.295}\,\Delta x_{t-4}\end{aligned}$$

$$R^2=0.300 \qquad \text{S.E.}=3.955, \tag{16}$$

which leads to a forecasting equation for real disposable income of the form

$$\begin{aligned}x_t = 1.920 + 1.478x_{t-1} - 0.478x_{t-2} + 0.309x_{t-3}\\ -0.604x_{t-4} + 0.295x_{t-5}.\end{aligned} \tag{17}$$

The one-period forecast error variance estimated over the sample period is $\hat{\sigma}_{uu}=15.642$.

The observation system for real disposable income was obtained from preliminary and revised data for the period 1965:I through 1974:IV. The results that were obtained are[3]

$$y_t = \underset{(2.23)}{19.641} + \underset{(59.2)}{0.955}\, x_t + v_t \tag{18}$$

$$v_t = \underset{(3.57)}{0.510}\, v_{t-1} + w_t. \tag{19}$$

The variance of w_t estimated over the sample period is $\hat{\sigma}_{ww}=9.098$. The observation system is completed by adding the revision equation (12) to the system.

[3]These parameter estimates were obtained via a two-step procedure. In the first step, least squares estimates of the parameters in (18) were obtained. The residuals from this equation were used to estimate the coefficient of v_{t-1} in (19). This estimate of the first-order correlation coefficient was used to transform the preliminary and revised data and a second regression using the transformed data was performed to obtain efficient estimates of the parameters in (18). These efficient estimates were used to define a corresponding set of residuals and a second estimate of the first-order correlation coefficient was obtained. The results at the end of this second round are shown in the text.

Forecasts of Revised Values of Disposable Income

The problem of forecasting the revised value of real disposable income is considered first. Three suboptimal forecasting procedures have been investigated in addition to the optimal predictor. The first of these simply ignores the preliminary data completely and uses the forecasting model and the revised observations. The predictor in this case is the conditional expectation of x_{t+1} given $x_{t-l}, x_{t-l-1},\ldots$, and is therefore the standard $(l+1)$-period predictor for an antoregressive process. The prediction error variance is computed according to the standard formula given, for example, in Box and Jenkins (1970, p. 128).

The second forecasting procedure simply treats the preliminary data as if they were not subject to measurement error. In particular, lagged values of disposable income in the forecasting equation are replaced by the available preliminary or revised observations so that the general form of the predictor is

$$\hat{x}_{t+1} = \phi_0 + \sum_{j=0}^{l-1} \phi_{j+1} y_{t-j} + \sum_{j=l}^{p} \phi_j x_{t-j}. \tag{20}$$

This predictor thus corresponds closely to current practice in macroeconometric forecasting. It is not difficult to verify that in general this predictor is biased and the prediction error variance depends on the characteristics of the forecasting model as well as on the properties of the observation system.

The third predictor that is considered here adjusts the preliminary observations for bias and serial correlation and uses this adjusted preliminary estimate as if it were not subject to measurement error. The general form of the predictor is[4]

$$\hat{x}_{t+1} = \phi_0 + \sum_{j=0}^{l-1} \phi_{j+1} \hat{y}_{t-j} + \sum_{j=l}^{p} \phi_j x_{t-j} \tag{21}$$

[4]These adjustments produce the best forecasts of the preliminary data given the observation system. An even better forecast can be obtained by using the observation system and the forecasting model. This is in fact what the optimal forecast procedure does.

where

$$\hat{y}_{t-j} = (y_{t-j} - \theta_0 - \hat{v}_{t-j})/\theta_1 \tag{22}$$

and

$$\hat{v}_{t-j} = \psi^{l-j} v_{t-l}. \tag{23}$$

The theoretical prediction error variances for the alternative prediction procedures are shown in table 1. These computations assume that the data were generated by the forecasting and observation models given in (17), (18), (19), and (12) with the disturbance variances that were estimated from the sample. They also assume, in the case of the optimal predictor shown in the fourth column of the table, that the disturbances are normally distributed.[5] The error variances are shown for values of the data revision lag ranging from zero to six quarters.

TABLE 1.—PREDICTION ERROR VARIANCES OF ALTERNATIVE PREDICTORS OF REAL DISPOSABLE INCOME

Data Revision Lag	Prediction Method			
	1	2	3	4
0	15.6	15.6	15.6	15.6
1	49.9	42.5	37.4	29.0
2	95.4	36.5	36.1	32.4
3	165.5	38.6	37.9	34.8
4	238.2	39.8	39.6	34.9
5	307.1	39.5	39.7	34.9
6	378.0	39.5	39.7	34.9

Several rather striking conclusions emerge from this table.

1. A one-period lag in the availability of revised data increases the prediction error variance substantially regardless of the predictor that is used. If revised data were available immediately, all four prediction methods would yield an identical prediction-error variance of 15.6. With just a one-quarter lag in the availability of revised data, the prediction-error variance nearly doubles. The increase in the error variance of the suboptimal predictors is even greater.
2. The advantage of the optimal predictor over the suboptimal methods is reduced by an increase in the revision lag. For a one-period lag, the optimal predictor has a decided advantage over the other methods. If the data revision lag exceeds one quarter, the advantage of the optimal predictor is less pronounced but nevertheless real.
3. Unless data revision occurs within three quarters, there is little to be gained from data revision from the point of view of prediction accuracy. For the predictions that use preliminary data (methods 2, 3, and 4), the prediction-error variance converges to a constant as the data revision lag increases without limit. This upper limit on the error variance is nearly attained when $l=3$. Thus, any increase in the delay in the availability of revised data beyond three quarters has little effect on prediction accuracy.
4. The forecast procedures that use preliminary data are superior to the autoregressive prediction that ignores the preliminary data. This is not surprising in this case because the error variance of the forecasting model exceeds the variance of the disturbances in the observation equation. This need not be the case in general, however.

These results are, of course, model dependent. However, if they are generally valid, they have rather interesting implications for the dissemination and use of preliminary and revised data.

The results shown in table 1 are expected or theoretical results that assume that the model is correctly specified and the parameter estimates are accurate. Whether these error variances would be observed in the actual use of the alternative forecasting methods depends on the accuracy of the models, including the assumption of normality of the disturbance terms. An experimental application of these alternative predictors was carried out for the period 1965:II through 1974:IV to check the accuracy of the theoretical calculations. Summary statistics of the prediction errors are shown in table 2.

These results are remarkably similar to the expected theoretical values of table 1. The differences that do emerge are rather interest-

[5] For the optimal predictor, equations (5) and (6) were solved sequentially from an arbitrary initial condition for P_1 until P_t converged to a constant matrix.

TABLE 2.—SUMMARY STATISTICS OF PREDICTION ERRORS FOR REAL DISPOSABLE INCOME 1965:II–1974:IV

Data Revision Lag		Prediction Method 1	2	3	4
	bias	−1.9	−5.6	0.2	−0.6
1	variance	77.4	38.4	31.4	31.4
	rmse[a]	12.6	8.3	5.6	5.7
	bias	−3.5	−4.1	0.4	−0.4
2	variance	121.0	34.8	32.5	25.0
	rmse	11.6	7.2	5.7	5.0
	bias	−6.4	−5.1	0.6	−0.3
3	variance	127.7	33.6	28.1	23.0
	rmse	13.0	7.8	5.3	4.8
	bias	−8.7	−3.7	0.2	−0.3
4	variance	161.3	34.8	31.4	24.0
	rmse	15.4	7.0	5.6	4.9
	bias	−10.1	−4.6	0.4	−0.3
5	variance	163.8	39.7	34.8	24.0
	rmse	16.3	7.8	5.9	4.9
	bias	−11.4	−4.6	0.5	−0.2
6	variance	158.9	41.0	36.0	25.0
	rmse	17.0	7.9	6.0	5.0

[a] Root-mean-square error.

ing. The advantage of the optimal predictor over the suboptimal predictors stands out even more clearly in these simulation results. In addition, the gain from a simple adjustment of the preliminary data is apparent in its reduction of both bias and variance. The use of unadjusted preliminary data leads to a bias on the order of $4 or $5 billion which is eliminated when the preliminary data are adjusted. It appears from these results that the expected gain in prediction accuracy from the use of the optimal predictor would indeed be realized.

Forecasts of Preliminary Values of Disposable Income

Forecasts are inevitably compared with published values to evaluate the accuracy of a forecasting model. In actual practice, recent prediction errors are frequently used to adjust the model before generating subsequent forecasts. Since these recent errors are calculated on the basis of preliminary data, forecasts of the preliminary data are desirable. Otherwise the error statistics will contain both prediction and measurement errors and may not be reliable measures of forecast accuracy. In other words, in order to ensure comparability, preliminary values should be compared with predicted preliminary values and revised values should be compared with forecasts of revised values.

It is of interest to consider a set of preliminary-data forecast procedures similar to those introduced previously for the prediction of revised data. In the case of the first predictor, the preliminary data are ignored completely in making the forecast. The second prediction method makes no distinction between preliminary and revised values so that predictions of preliminary and revised values are identical. The third prediction method explicitly recognizes the distinction between preliminary and revised data. Predictions of the preliminary data are generated by substituting forecasts of the revised data obtained by the third method into the observation system. The fourth optimal forecast procedure substitutes the optimal forecast of the revised value into the observation equation.

The prediction error variances for these alternative forecast procedures are shown in table 3 for different values of the data revision lag ranging from zero to six. Several interesting conclusions emerge from this table.

1. It always pays to use preliminary data in the forecast procedure rather than ignore them. Once again, this is because the error variance of the observation equation is smaller than the error variance of the forecasting equation. This need not be true in general.
2. The naive use of preliminary data is inferior to the more sophisticated techniques that adjust the preliminary data for bias and serial correlation. In particular, the optimal predictor yields a noticeable improvement in forecast accuracy over the suboptimal predictors.
3. The three prediction methods that use preliminary data are able to predict preliminary values more accurately than they are able to predict revised values.

This last result is especially interesting. It suggests that a model will appear to yield more accurate forecasts if the predictions of the model are compared with preliminary data than if the forecasts are compared with revised data.

TABLE 3.—PREDICTION ERROR VARIANCES OF ALTERNATIVE PREDICTORS OF PRELIMINARY VALUES OF REAL DISPOSABLE INCOME

Data Revision Lag	Prediction Method			
	1	2	3	4
0	27.9	27.9	23.4	23.4
1	62.2	36.3	31.9	28.6
2	107.7	33.3	31.9	29.2
3	177.8	34.5	34.1	30.1
4	250.5	36.6	35.9	30.3
5	319.4	36.0	36.0	30.3
6	390.3	36.0	36.0	30.4

IV. Forecasts of Personal Consumption Expenditure

Prediction of disposable income provides an interesting example of the use of preliminary data in econometric forecasting. However, the forecasting model is overly simple and does not provide the true flavor of the optimal forecasting procedure. It is therefore of interest to consider a slightly more complicated situation involving more than a single equation in the forecasting model. The model that will now be investigated consists of an aggregate consumption function together with the disposable income equation.

A consumption function was estimated by ordinary least squares using the most recently revised data for the period 1954:I–1974:IV. The results of this analysis are

$$c_t = 5.995 + 0.865 x_t + \epsilon_t \tag{24}$$

$$\epsilon_t = 0.694 \epsilon_{t-1} + \eta_t \tag{25}$$

where c_t denotes real personal consumption expenditure and x_t denotes real disposable income. The variance of η_t estimated from the sample is $\hat{\sigma}_{\eta\eta} = 9.251$. The observation system for personal consumption expenditure is

$$z_{1t} = -1.085 + 1.000 c_t + \nu_t \tag{26}$$

$$\nu_t = 0.676 \nu_{t-1} + \kappa_\tau \tag{27}$$

$$z_{2t} = c_{t-l} \tag{28}$$

with $\hat{\sigma}_{\kappa\kappa} = 3.753$. Both of these systems were estimated using the two-stage procedure described previously.

The forecast system now consists of equations (17), (24), and (25) and the observation system is composed of (18), (19), (12), (26), (27), and (28). The theoretical prediction error variances for this system are shown in table 4 for the four types of predictors that were introduced in the previous section. With the exception of the optimal predictor, the entries for disposable income error variances are the same in table 4 as the corresponding entries in table 1. The optimal predictor for the personal consumption expenditure-disposable income system has a smaller prediction error variance for disposable income than before. For example, for $l = 1$, the error variance is 29.0 for the disposable income equation alone and 25.5 for the consumption-income system. The reason for this reduction in variance is that the consumption function and preliminary observations on consumption expenditure provide additional information about disposable income that can be used to improve the forecast. A comparison of the corresponding entries in tables 1 and 4 indicates that the system approach leads to a significant reduction in prediction error. This is an extremely important result in the sense that it provides a strong motivation for an extension of the optimal forecast methods to realistic economic forecasting systems. Even further gains in forecast accuracy can be expected in more complex, interdependent systems.

The system effect is perhaps the most striking result that emerges from this table but there are several other points of interest.

TABLE 4.—PREDICTION ERROR VARIANCES OF ALTERNATIVE PREDICTIONS OF REAL DISPOSABLE INCOME AND REAL CONSUMPTION EXPENDITURE

Data Revision Lag		Prediction Method			
		1	2	3	4
1	c	50.9	29.9	27.4	25.5
	x	49.9	42.5	37.4	25.5
2	c	86.9	28.5	27.9	26.4
	x	95.4	36.5	36.1	27.7
3	c	140.3	29.1	28.7	27.2
	x	165.3	38.6	37.9	29.5
4	c	195.0	31.0	30.5	27.5
	x	238.2	39.8	39.6	29.6
5	c	246.6	30.5	30.6	27.6
	x	307.1	39.5	39.7	29.6
6	c	299.6	30.5	30.7	27.7
	x	378.0	39.5	39.7	29.6

1. The optimal predictor results in significant gains in prediction accuracy for both consumption and disposable income.
2. If revised data are not available within two or at most three quarters, revisions are of little use in forecasting.
3. There are substantial gains associated with the use of preliminary data even if they are used in a suboptimal way.

These results document the importance of measurement error in preliminary data and indicate clearly that intelligent use of preliminary data would be expected to result in a meaningful reduction in prediction error variances.

V. Concluding Remarks

In the construction of ex ante econometric forecasts, the distinction between preliminary and revised data is typically ignored. This results in an unnecessarily large variance of forecast errors. The optimal procedure analyzed in this article automatically incorporates the distinction between preliminary and revised data in the forecasting procedure to improve prediction accuracy.

In order to determine the potential gain in forecast accuracy that could be expected from the optimal use of preliminary data, several empirical models were analyzed. In the prediction of disposable income alone, the naive use of preliminary data results in forecast error variances that exceed the variance of the optimal predictor by 11% to 47% depending on the length of the data revision lag. When disposable income and personal consumption expenditure are forecast simultaneously the margin of improvement is even larger. In this case the naive prediction error variance exceeds the variance of the optimal predictor of disposable income by 31% to 67%. The potential gains from the optimal use of preliminary data are by no means inconsequential!

As in all studies of this type, the quantitative results depend on the models that are used. Even with the simple models investigated here, substantial gains in forecast accuracy are possible. It remains to be seen if even larger improvements can be achieved with more complex econometric forecasting models.

REFERENCES

Astrom, Karl J., *Introduction to Stochastic Control Theory* (New York: Academic Press, 1970).

Box, George E. P., and Gwilym M. Jenkins, *Time Series Analysis: Forecasting and Control* (San Francisco, Holden-Day, 1970).

Cole, Rosanne, *Errors in Provisional Estimates of Gross National Product* (New York: National Bureau of Economic Research, 1969a).

———, "Data Errors and Forecasting Accuracy," chapter 2 in Jacob Mincer (ed.), *Economic Forecasts and Expectations: Analyses of Forecasting Behavior and Performance* (New York: National Bureau of Economic Research, 1969b).

Denton, Frank T., and John Kuiper, "The Effect of Measurement Errors on Parameter Estimates and Forecasts: A Case Study Based on the Canadian Preliminary National Accounts," this REVIEW 47 (May 1965), 198–206.

Denton, Frank T., and Ernest H. Oksanen, "A Multi-Country Analysis of the Effects of Data Revisions on an Econometric Model," *Journal of the American Statistical Association* 67 (June 1972), 286–291.

McDonald, John, "On the Insensitivity of the Autoregressive Moving Average Representations of Some Australian Quarterly Time Series," *Econometrica* 44 (Nov. 1976), 1277–1287.

Morgenstern, Oskar, *On the Accuracy of Economic Observations* (Princeton: Princeton University Press, 1963).

Stekler, H. O., "Data Revisions and Economic Forecasting," *Journal of the American Statistical Association* 62 (June 1967), 470–483.

Zellner, Arnold, "A Statistical Analysis of Provisional Estimates of Gross National Product and Its Components, of Selected National Income Components, and of Personal Savings," *Journal of the American Statistical Association* 52 (Mar. 1958), 54–65.